Python Quiz 4 - While Loop Practice

Instructions:

Trace each program carefully.

- If the program terminates normally, write its exact output.
- If it repeats forever, identify the infinite loop and explain why.
- If an error occurs, identify the error and explain when it occurs.

Question 1 - Trace the Output

What is displayed when the following code is executed?

```
round_number = 1
tokens = 3

while round_number <= 4:
    if tokens < 7:
        tokens += round_number
    else:
        tokens -= 2

    round_number += 1

print(tokens, round_number)
```

Expected output:

Question 2 - Trace the Output

What is displayed when the following code is executed?

```
measurements = [3, 8, 2, 5, 7]
location = len(measurements) - 1
score = 0

while location >= 0:
    if measurements[location] > 4:
        score += measurements[location]
        location -= 2
    else:
        score += 1
        location -= 1

print(score, location)
```

Expected output:

Question 3 - Trace the Output

What is displayed when the following code is executed?

```
numbers = []
current = 2

while current < 9:
    numbers.append(current + 1)
    current += 3

print(numbers, current)
```

Expected output:

Question 4 - Trace the Output

What is displayed when the following code is executed?

```
word = "CAMERA"
place = 0
result = ""

while place < len(word):
    result += word[place]

    if word[place] == "A":
        place += 2
    else:
        place += 1

print(result, place)
```

Expected output:

Question 5 - Trace the Output

What is displayed when the following code is executed?

```
message = "PYTHON"
place = 4
result = ""

while place > 0:
    result += message[place]
    place -= 2

print(result, place)
```

Expected output:

Question 6 - Infinite Loop Analysis

Study the following code:

```
route = ["A", "B", "C"]
location = 0
visits = 0

while location < len(route):
    visits += 1

    if route[location] == "B":
        location = 0
    else:
        location += 1

print(visits, location)
```

1. Does the loop terminate?
2. What values does location repeatedly take?
3. Is the print() statement ever executed?
4. Briefly explain why the loop does or does not terminate.

Answers and explanation:

Question 7 - Multiple Choice

What happens when the following code is executed?

```
names = ["Lina", "Omar", "Chen"]
place = 0
initials = ""

while place <= len(names):
    initials += names[place][0]
    place += 1

print(initials)
```

(a) The code displays LOC.	**(b)** The code displays LO.
(c) An IndexError occurs before the print() statement is reached.	**(d)** Nothing is displayed because the loop repeats forever.